

Md. Shahriyar Mahmud Tasrif

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About

I am a final-year undergraduate student in Electrical and Electronic Engineering at BUET, majoring in **Electronics and Photonics**. My primary research interests lie in **Spintronic devices**, **Non-Volatile Memory architectures**, and **Neuromorphic hardware design**; my current thesis work centres on a hybrid spintronic-CMOS pulse width modulation generator operating in the GHz regime. Broadly, my academic interests encompass Solid State Devices, Compound Semiconductors, Optoelectronics, VLSI Design (Analog & Digital), and Computer Architecture. My goal is an academic career where classroom teaching and active research directly support each other.

Education

Bangladesh University of Engineering and Technology (BUET) <i>B.Sc. in Electrical and Electronic Engineering</i> Major: Electronics and Photonics ◦ CGPA: 3.93/4.00 (Top 6% of class) ◦ Recent Performance: Perfect GPA 4.00 in the last two consecutive terms. ◦ All coursework completed; thesis defended June 2026 — final results pending.	<i>Dec 2021 – Jun 2026</i> <i>Up to 7th Semester</i>
Dr. Mahbubur Rahman Mollah College <i>Higher Secondary Certificate (HSC)</i> ◦ GPA: 5.00/5.00	<i>2020</i>
Gazipur Khan Govt. High School and College <i>Secondary School Certificate (SSC)</i> ◦ GPA: 4.94/5.00	<i>2018</i>

Teaching Interests

Device Physics & Materials: Solid State Devices, Compound Semiconductor Devices, Semiconductor Physics, Optoelectronics, and Electrical Properties of Materials.
Circuits & Systems: VLSI Circuits and Design, Analog Integrated Circuit Design, Digital Electronics, and Microprocessor & Embedded Systems.

Honors & Awards

- **Dean's List Award**, BUET *(2022–2025)*
 - Awarded for maintaining a GPA of 3.75 or above in two regular terms of an academic year.
 - Received the award in each academic year to date.
- **University Merit Scholarship**, BUET *(2022–2025)*
 - Awarded to the top 40 students in the Engineering Faculty based on term examination results.
 - Received the scholarship in each academic year.
- **Candidate for Bachelor's Degree with Honors**
 - Tracking to graduate with Honors (awarded for overall CGPA of 3.75 or better).
- **Champion**, Bangladesh Mathematical Olympiad (Regional Level).
- **Champion**, DMRC Science Club Mathematics Olympiad.

Research Experience

Voltage Controlled High-Frequency Pulse Width Modulation Generation based on Spin Hall Nano-Oscillator <i>Undergraduate Thesis — Supervisor: Dr. Orchi Hassan, Dept. of EEE, BUET</i> <i>Thesis defended June 2026; final results pending.</i>	<i>March 2025 – June 2026</i>
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- Designed a hybrid spintronic-CMOS PWM generator using a field-free Spin Hall Nano Oscillator (SHNO) driven by spin-orbit torque, replacing conventional high-speed CMOS oscillators.
- Simulated and compared two circuit architectures (7T-1MTJ single-pFET and CMOS transmission-gate variants), using a macrospin LLGS MTJ model and PTM FinFET cards.
- Demonstrated GHz-range oscillation with 6–95% duty cycle tuning and sub-50 μ W power consumption — the first PWM generator design simultaneously achieving these benchmarks.

Spintronic Non-Volatile Memory: Device Physics and Circuit Design

March 2025 – Present

Undergraduate Researcher — Supervisor: [Dr. Orchi Hassan](#), Dept. of EEE, BUET

- Investigating Spin-Transfer Torque (STT) and Spin-Orbit Torque (SOT) switching mechanisms in Magnetic Tunnel Junctions (MTJs) for application in next-generation Non-Volatile Memory (NVM) architectures.

Industrial Attachment

Neural Semiconductor Limited (NSL), Uttara, Dhaka

November 2025

- Worked on the full IC design cycle across Analog, Digital VLSI, and RFIC domains.
- Practised area-efficient layout, impedance matching, and power-aware RTL design.
- Performed DRC/LVS-based physical verification and design closure using industry EDA tools.

Selected Projects

16-bit Binary Square Rooter

VLSI Laboratory

Tools: Verilog, SystemVerilog, Cadence Virtuoso, Cadence Innovus [🔗 Project Files](#)

- Designed a fully synchronous 16-bit integer square root unit (shift-and-subtract, 8-cycle latency, no multipliers) through the complete RTL-to-GDS flow with layered testbench verification and synthesis-level reporting.

Low Noise RF Amplifier for Wireless Communication

Analog Integrated Circuit Laboratory

Tools: Cadence Virtuoso, Cadence Spectre, gpd090 CMOS [🔗 Project Files](#)

- Designed a 1.8 GHz common-source LNA with inductive source degeneration, analytically optimised for minimum noise figure and 50 Ω input matching, verified via S-parameter and noise simulations.

Chromabot: Embedded Line Follower with Multi-Color Recognition

Microprocessor & Embedded Systems Laboratory

Tools: STM32 ARM Cortex-M, C firmware, I2C, TCS34725 RGB sensors [🔗 Project Files](#)

- Built an autonomous robot interpreting colour-encoded commands in real time via three I2C RGB sensors; developed the full firmware stack including colour classification logic and PWM motor control.

Aging-Induced Spectral Variation Study of TFT Display Devices

Optoelectronics Laboratory

Tools: ESP32, Python, MATLAB, Webcam [🔗 Project Files](#)

- Built a low-cost DIY spectrometer to track daily spectral degradation of a TFT-LCD panel over 22 days; fitted first-order exponential decay models and quantified per-channel brightness loss across RGB bands.

SunSeeker: Dual-Axis Adaptive Solar Tracking System

Control Systems Laboratory

Tools: Arduino, servo actuators, LDR sensors [🔗 Project Files](#)

- Implemented a closed-loop dual-axis solar tracker computing real-time panel orientation from solar declination and hour-angle equations, with servo-based feedback for continuous sun alignment.

Electric Elevator Control System

Digital Logic Design Laboratory

Tools: Verilog, System Verilog, Quartus Altera, FPGA [🔗 Project Files](#)

- Designed a digital elevator controller with floor-request handling, direction arbitration, and door-control logic using combinational and sequential digital circuits, implemented on FPGA.

Smart Power Supply: Variable DC with Adaptive Battery Charging

Power Electronics Laboratory

Tools: Microcontroller, Proteus, LTSpice [🔗 Project Files](#)

- Designed a variable DC supply with adaptive battery charging, implementing automatic voltage regulation and current-limiting protection circuits on a hardware prototype.

Behavioral Model of an Operational Amplifier from Datasheet

Electronic Circuits Laboratory

Tools: LTSpice [🔗 Project Files](#)

- Constructed a passive-element behavioral model of the ADA-4004 op-amp, extracting gain, I/O resistance, bandwidth, bias current, and offset parameters directly from the manufacturer datasheet.

Relevant Coursework

- Major Courses:** (*Electronics, Photonics, VLSI, & Digital Systems*)
Compound Semiconductor Devices, Solid State Devices, Nano-electronics & Nanotechnology, Optoelectronics, Semiconductor & Nano Devices, Fabrication & Processing Technology, Electrical Properties of Materials, VLSI Circuits and Design, Analog Integrated Circuit Design, Microprocessor & Embedded Systems, Digital Electronics, Engineering Electromagnetics.
- Breadth Courses:** (*Power, Communication & Circuits*)
Digital Signal Processing, Continuous Signals and Linear Systems, Control Systems, Communication Systems (I & II), Power Electronics, Power Systems, Energy Conversion (I & II), Electrical & Electronic Circuits (4 courses).

Skills

- Languages & Scripting:** Python, Verilog, SystemVerilog, MATLAB, C, C++.
- EDA & Simulation:** HSPICE, Cadence Virtuoso, Cadence Spectre, ADS, Quartus Altera, Vivado Xilinx, Microwind, DSCH, Proteus, LTSpice, OrCAD PSpice, PSAF.
- Hardware:** FPGA (Xilinx/Altera), Microcontrollers (ARM — STM32/Raspberry Pi, AVR — Arduino).
- Spoken Languages:** Bangla (Native), English (Fluent), Hindi (Conversational), Spanish (Basic).

Extracurricular Activities

- Academic Team Member** — *Bangladesh Mathematical Olympiad (BdMO)* *Jan 2022 – Present*
◦ Mentored students and assisted in organizing national-level mathematics competitions over four consecutive years.
- Co-Organizer, BUET EEE Day 2026** — *Dept. of EEE, BUET* *2026*
◦ Arranged a University/College-level Integration Bee competition, managed the EEE Olympiad, and coordinated the departmental cultural programme.

References

Dr. Orchi Hassan

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(*Thesis Supervisor*)

Dr. Muhammad Anisuzzaman Talukder

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